Lecture 4

Recap

1. For a function $f \in L^1([0,T])$

(i) $|c_n| \leq \frac{1}{T}\int_0^T |f(t)|dt$ (ii) $\lim_{n\to\pm\infty} c_n = 0$.

2. If f is k-times continuously diff^{bl}

$$|c_n| \leq M/|n|^k.$$

3. Dirichlet's criterion

If $f_+'(t)$ & $f_-'(t)$ exist, then

$$\sum_{n=-\infty}^{\infty} c_n e^{in\Omega t} = \frac{f(t^+)+f(t^-)}{2}$$

discontinuous at t: Fourier series equals the average of the left & right limits.

continuous at t: Fourier series equals the value of f at t

1.7: L^2 theory & Parseval's thm

Let f, g be two T-periodic f^n. Then for any $\alpha, \beta \in \mathbb{C}$ we have

$$\begin{aligned}(\alpha f + \beta g)(t+T) &= \alpha f(t+T) + \beta g(t+T) \\ &= \alpha f(t) + \beta g(t) \\ &= (\alpha f + \beta g)(t)\end{aligned}$$

$\Rightarrow \alpha f + \beta g$ is also T-periodic.

This gives the set of T-periodic f^n's the structure of a vector space. Here the functions play the role of our 'vectors'. This begs the question: can we use linear algebra to study T-periodic f^n's?

First recall that vector spaces can be given an inner product. For example, in $\mathbb{C}^n$ the standard inner product is

$$\vec{u} \cdot \vec{v} = \sum_{k=1}^{n} u_k \overline{v}_k.$$

↪ Encodes important geometric data

1. Squared length:
$$|\vec{u}|^2 = \vec{u} \cdot \vec{u}$$

2. Orthogonality
$\vec{u} \cdot \vec{v} = 0 \Rightarrow$ extends idea that lines tangent to $\vec{u}$ & $\vec{v}$ meet at right angles

In fact, we have the Cauchy-Schwarz inequality

$$|\vec{u}\cdot\vec{v}| \leq |\vec{u}||\vec{v}|,$$

this implicitly defines an angle between $\vec{u}$ & $\vec{v}$!

Using the inner product we can also define an orthonormal basis (ON) $\{\vec{u}_k : k=1,\dots,n\}$ as a set of vectors that span the whole space & such that

$$\vec{u}_i \cdot \vec{u}_j = \begin{cases} 1, & \text{if } i=j, \\ 0, & \text{if } i \neq j \end{cases}$$

Then any vector $\vec{v} \in \mathbb{C}^n$ can be written as

$$\vec{v} = \sum_{k=1}^{n} \alpha_k \vec{u}_k.$$

The constants can be easily obtained:

$$\vec{u}_m \cdot \vec{v} = \sum_{k=1}^{n} \alpha_k (\vec{u}_m \cdot \vec{u}_k)$$

($\vec{u}_m \cdot \vec{u}_k$: non-zero iff $m=k$.)

$$= \alpha_m.$$

All of this is similar to Fourier series:

roughly speaking:
- $\vec{v}$ is some T-periodic f
- $\vec{u}_k$ are complex exponentials
- α_k are Fourier coefficients

To make this analogy more precise, we can define the inner product

$$\langle f, g \rangle = \frac{1}{T} \int_0^T f(t) \overline{g(t)}\, dt.$$

⇝ Just like before, but vectors are replaced by f^n's indexed by $[0, T)$ and

$$\Sigma \longrightarrow \int_0^T dt.$$

If we now restrict to the class of f^n's such that

$$\|f\|_2^2 = \frac{1}{T} \int_0^T |f(t)|^2 dt < \infty,$$

we get the space $L^2([0,T])$ $(L^2(\mathbb{T}))$. Equipping this space w/ the inner product above makes it an inner product space. It is easy to show

$$\langle e^{im\Omega t}, e^{in\Omega t} \rangle = \begin{cases} 1, & \text{if } m = n, \\ 0, & \text{if } m \neq n. \end{cases}$$

⇝ Just like an orthonormal basis!

(Mathematically, we're being sketchy but that's OK.)

Using this idea, we can think of the Fourier series of an L^2 function as an expansion into an ON basis w/ coefficients

$$c_m \doteq \langle f(t), e^{im\Omega t} \rangle$$
$$= \frac{1}{T} \int_0^T f(t) e^{-im\Omega t} \, dt.$$

Similar statements are true for the trig. expansion, its just a basis change.

The quantity $\|f\|$ is the analogue of the squared length, we have the following neat result.

Thm 1.2: If $f \in L^2([0,T])$, then

$$\frac{1}{T} \int_0^T |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2$$
$$= \frac{|a_0|^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2).$$

⇝ This is Parseval's theorem, it is the analogue of Pythagorus's thm for Fourier series.

Proof:

$$\|f\|^2 = \frac{1}{T} \int_0^T f(t) \overline{f(t)} \, dt$$

$$= \frac{1}{T}\int_0^T f(t)\left[\sum_{n=-\infty}^{\infty} \overline{c_n}\, e^{-in\Omega t}\right] dt.$$

Assuming we can exchange the integral w/ the summation, we get

$$\|f\|^2 = \sum_{n=-\infty}^{\infty} \frac{\overline{c_n}}{T} \underbrace{\int_0^T f(t)\, e^{-in\Omega t}\, dt}_{T c_n = T\langle f, e^{in\Omega t}\rangle}$$

$$= \sum_{n=-\infty}^{\infty} \overline{c_n} c_n$$

$$= \sum_{n=-\infty}^{\infty} |c_n|^2.$$

The result for a_n & b_n follow by writing

$$c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \frac{a_n + ib_n}{2}$$

$$c_0 = a_0/2.$$

Then

$$\|f\|^2 = |c_0|^2 + \sum_{n=1}^{\infty} (|c_n|^2 + |c_{-n}|^2)$$

$$= \frac{|a_0|^2}{4} + \frac{1}{4}\sum_{n=1}^{\infty} \underbrace{\left[|a_n|^2 + |b_n|^2 + |a_n|^2 + |b_n|^2\right]}_{\text{extra terms cancel out!}}$$

$$= \frac{|a_0|^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)$$

□

Example 1.10

i. The sawtooth wave belongs to L^2:

$$\|f\|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t^2\,dt$$

$$= \frac{1}{2\pi}\left[\frac{t^3}{3}\right]_{-\pi}^{\pi} = \frac{\pi^2}{3}.$$

We saw that the Fourier series of f is given by

$$f(t) \sim \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{(-1)^{n+1}}{in} e^{int}.$$

Parseval's theorem then states

$$\frac{\pi^2}{3} = \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{2}{n^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

ii. Recall that the f^n

$$f(t) = t^2, \qquad -1 \leq t < 1$$

has Fourier series

$$f(t) \sim \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos n\pi t$$

The function belongs to L^2:

$$\frac{1}{2}\int_{-1}^{1} |t^2|^2\,dt = \frac{1}{2}\int_{-1}^{1} t^4\,dt$$

$$= \frac{1}{10}\left[t^5\right]_{-1}^{1} = \frac{1}{5}.$$

The trig. form of Parseval's th^{m} then gives

$$\frac{1}{5} = \frac{1}{9} + \frac{1}{2}\cdot\frac{16}{\pi^4}\sum_{n=1}^{\infty}\frac{1}{n^4}$$

$$\Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^4} = 2\cdot\frac{\pi^4}{16}\left(\frac{1}{5}-\frac{1}{9}\right)$$

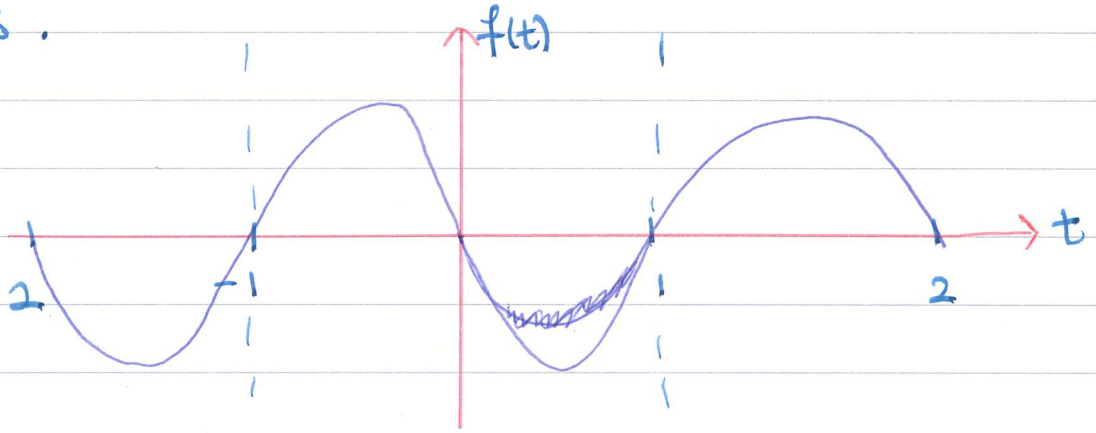
$$= \frac{\pi^4}{90}.$$

Long example

Lets bring together everything we learned thus far by finding the Fourier series of

$$f(t) = t^3 - t,$$

on $-1 \le t < 1$. Then we will use Parsevals & Dirichlet's criterion to get series results.



This is an odd f^n, so we immediately have

$$a_n = 0, \qquad \forall n.$$

The sine coefficients are:

$$b_n = \int_{-1}^{1} (t^3 - t)\sin(n\pi t)\,dt$$

$$= \left[\frac{-(t^3-t)\cos(n\pi t)}{n\pi}\right]_{-1}^{1} + \frac{1}{n\pi}\int_{-1}^{1}(3t^2-1)\cos(n\pi t)\,dt$$

(Both vanish)

$$= \frac{1}{(n\pi)^2}\left[(3t^2-1)\sin(n\pi t)\right]_{-1}^{1} - \frac{1}{(n\pi)^2}\int_{-1}^{1} 6t\sin(n\pi t)\,dt$$

($\sin n\pi = 0 \Rightarrow$ vanish!)

$$= +\frac{1}{(n\pi)^3}\left[6t\cos(n\pi t)\right]_{-1}^{1} + \underbrace{\frac{-6}{(n\pi)^2}\int_{-1}^{1}\cos(n\pi t)\,dt}$$

(seen before, it vanishes!)

$$= \frac{12(-1)^n}{n^3\pi^3}.$$

$$\Rightarrow f(t) \sim \frac{12}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\sin(n\pi t).$$

To check Dirichlet's criterion, we only need to check $t=1$. Everywhere else it is smooth (within a period).

$$f_-'(1) = \lim_{h\to 0^-} \frac{(1+h)^3-(1+h)-0}{h}$$

$$= \lim_{h\to 0^-} \frac{h^3+3h^2+3h+1-(1+h)}{h}$$

$$= \lim_{h\to 0^-} \frac{h^3+3h^2+2h}{h} = 2.$$

$$f_+'(1) = \lim_{h\to 0^+} \text{exercise!} = 2$$

↳ on $[1,3)$

$f(t) = (t-2)^3 - (t-2)$

$\Rightarrow$ Differentiable at $t=1$.

Hence,

$$f(t) = \frac{12}{\pi^3}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}\sin(n\pi t).$$

At $t = 0, \pm 1$, both sides vanish as expected. If we take $t = 1/2$, we get

$$\left(\frac{1}{2}\right)^3 - \frac{1}{2} = \frac{12}{\pi^3}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}\sin\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}\sin\left(\frac{n\pi}{2}\right) = \frac{\pi^3}{12}\left[-\frac{3}{8}\right]$$

$$= \frac{-\pi^3}{32}$$

Finally, f belongs to L^2:

$$\frac{1}{2}\int_{-1}^{1}(t^3-t)^2\,dt = \frac{1}{2}\int_{-1}^{1}(t^6-2t^4+t^2)\,dt$$

$$= \frac{1}{2}\left[\frac{t^7}{7} - \frac{2t^5}{5} + \frac{t^3}{3}\right]_0^1 \cdot 2$$

even!

$$= \frac{1}{7} - \frac{2}{5} + \frac{1}{3} = \frac{8}{105}.$$

Parseval's th^{m} then states

$$\frac{8}{105} = \frac{1}{2}\sum_{n=1}^{\infty}\frac{144}{n^6\pi^6}$$

$$\Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^6} = \frac{\pi^6}{945}.$$